

Linear Regression

Wednesday, June 24, 2026 7:37 PM

Non linear pattern ~~not~~ ~~not~~ ~~not~~

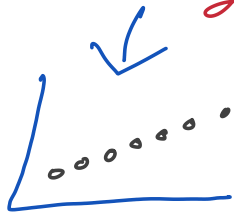
input & output
Linear Relation ~~not~~ ~~not~~ ~~not~~

LR

$$y = mx + c \text{ (Simple)}$$

$$y = m_1x_1 + m_2x_2 + \dots + m_nx_n + c \text{ (Multi)}$$

model best line
~~not~~ ~~not~~ ~~not~~



calculate the accuracy of Regression model

Matrix:

(i) MAE = measure avg size of error

(ii) MSE = Cost function $\rightarrow$ our goal is to minimize the MSE.
 $\rightarrow$ Lower cost = better line.

(iii) RMSE $\rightarrow$ how well observed data point match the expected value

(iv) R^2 = R Squared $\rightarrow$ mentioned how perform the model

$\rightarrow$ Range (0-1) $\rightarrow$ $R=1 \rightarrow$ best
 $\rightarrow$ $R=0 \rightarrow$ bad
 $\rightarrow$ $R=0.5 \rightarrow$ moderate

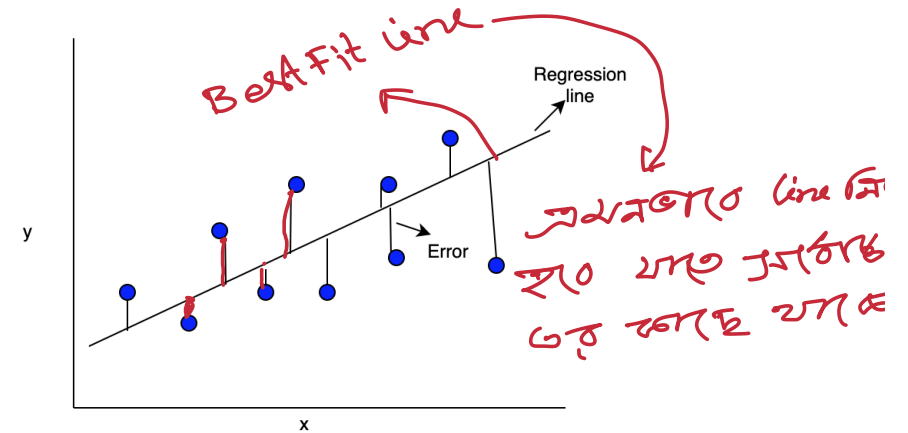
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Actual
Predicted

Evaluation Metrics

A variety of evaluation measures can be used to determine the strength of any linear regression model. These assessment metrics often give an indication of how well the model is producing the observed outputs.

- **Mean Squared Error (MSE):** Measures the average squared difference between actual and predicted values to avoid cancellation of errors.
- **Mean Absolute Error (MAE):** Calculate the accuracy of a regression model. MAE measures the average absolute difference between the predicted values and actual values.
- **Root Mean Squared Error (RMSE):** Square root of the residuals variance is RMSE. It describes how well the observed data points match the expected values or the model's absolute fit to the data.
- **R-Squared:** Indicates how much variation the model explains. Its value is typically between 0 and 1, but it can be negative if the model performs worse than a simple baseline model (e.g., predicting the mean).
- **Adjusted R-square:** Measures the proportion of variance explained by the model while adjusting for the number of predictors and penalizing irrelevant features.



Applications

- **Real Estate Price Prediction:** Estimate house prices using features such as location, area, and number of rooms.
- **Sales and Demand Forecasting:** Predict future sales and product demand based on historical trends and seasonal data.
- **Financial Analysis:** Analyze market trends, stock prices, and the impact of economic factors like inflation and interest rates.
- **Healthcare and Medical Research:** Predict disease progression, patient outcomes, and treatment effectiveness using medical data.
- **Marketing and Advertising Analysis:** Measure how advertising campaigns and promotional spending influence sales and customer engagement.

Regularization Techniques

- **Lasso Regression:** Regularizes a linear regression model, it adds a penalty term to the linear regression objective function to prevent overfitting.
- **Ridge regression:** Adds a regularization term to the standard linear objective to prevent overfitting by penalizing large coefficient in linear regression equation. It is useful when the dataset has multicollinearity where predictor variables are highly correlated.
- **Elastic Net Regression:** Hybrid regularization technique that combines the power of both L1 and L2 regularization in linear regression objective.

